

EDUCATION

PES University Bengaluru, India
B.Tech in Computer Science & Engineering 2022 - 2026

EXPERIENCE

Wayfair Bengaluru, India
Software Engineer Intern Jan '26 - Jun '26

Java, Spring Boot, FastAPI, React.js, Python, RAG, ChromaDB, LLMs, Docker, Kafka, Microservices, BigQuery

- Engineered **Noktra**, an AI-assisted onboarding platform, with hands-on ownership across the **full-stack application and AI pipeline**, automating complex multi-repository supplier onboarding from 6-8 weeks to hours, a **98% reduction** in turnaround time.
- Built a **RAG-powered generation engine** using **ChromaDB** and **LLMs** to generate SQL, YAML, and Terraform artifacts, reducing manual configuration effort by **94%**, while developing **React/FastAPI workflows** with rule-based validation, gate checks, and conflict detection to make AI-generated outputs reliable within real engineering workflows and reduce review rework by **90%**. Delivered an automated **review-assistance layer** with real-time similarity checks and actionable fix suggestions, accelerating cross-team PR approvals by **99%**.

Indian Space Research Organisation - UR Rao Satellite Centre (URSC) Bengaluru, India
Summer Intern Jun '24 - Jul '24

C, MATLAB, RRT*, Image Processing, Path Planning, Research and Development

- Engineered a **rover navigation system** combining modified **RRT* path planning** and lunar surface image processing in **MATLAB**, incorporating rover kinematic constraints, goal bias, and collision detection to generate efficient and feasible trajectories while using **Canny edge detection**, **subregion analysis**, and **boundary extraction** to identify and map hazardous terrain obstacles.
- Integrated the planning and perception pipelines into a unified navigation framework and validated the system through end-to-end **2D simulations**, achieving **95%+ navigation success** across diverse scenarios while demonstrating robust, collision-free trajectory generation.

PUBLICATIONS

An Integrated Platform for Adaptive Exam Preparation and Dynamic Question Generation in Higher Education

V. Marathe, R. K. Thomas, P. Sharma, S. K. Katenahalli, & Manju

2026 6th International Conference on Electrical, Computer and Energy Technologies (ICECET), Rome, Italy

[IEEE Xplore]

PROJECT HIGHLIGHTS

PESUprep - Adaptive Learning & AI-Powered Question Gen. Platform 2025

React.js, Node.js, MongoDB, Python, Hugging Face, BERT, OAuth 2.0

- Selected for the 49th series Student Projects Programme (SPP) by the **Karnataka State Council for Science and Technology (KSCST)**
- Built a full-stack adaptive learning platform with faculty and student dashboards and secure **OAuth 2.0** role-based access, complemented by an **NLP pipeline** using TF-IDF, summarization, and concept extraction to process PDFs, DOCX, and PPTX files into structured learning concepts. Developed an AI-powered question generation engine using **DistilGPT2** and **fine-tuned LLMs** to generate context-aware MCQs, descriptive, and fill-in-the-blank questions, and applied a **BERT-based Bloom's Taxonomy classifier** to categorize questions by cognitive level and difficulty.

Vocal Intelligent Platform - AI-Powered Acoustic Analysis & Coaching 2026

Next.js, TypeScript, Python, FastAPI, Librosa, Llama 3 (Groq API), PostgreSQL, MinIO, Docker

- Architected a **full-stack microservices platform** for evidence-based vocal coaching, orchestrating **4 services** to process audio and generate feedback in under **2 seconds**.
- Designed a Python **Digital Signal Processing (DSP)** engine using Librosa and Parselmouth to extract acoustic features including pitch stability, jitter, and shimmer with **>95% accuracy** compared to clinical voice analysis tools, and integrated **Llama 3 via Groq API** to translate complex DSP outputs into conversational coaching feedback with average inference latency below **500ms**.
- Containerized the platform using **Docker** and **Docker Compose**, reducing environment setup time by **90%** and enabling the architecture to support **100+ concurrent vocal processing requests**.

SKILLS

- Languages:** Java, Python, C++, JavaScript, TypeScript, SQL
- Software:** Spring Boot, FastAPI, React.js, Node.js, REST APIs, Microservices, Git
- AI/ML:** PyTorch, scikit-learn, Hugging Face, BERT, NLP, LLMs, RAG, ChromaDB
- Cloud/Data:** Docker, Kubernetes, Terraform, Kafka, CI/CD, GCP/GKE, PostgreSQL, MongoDB
- Core/Professional:** DSA, OOP, OS, Computer Networks, Technical Writing, Research & Analysis, Problem Solving